

# Lucas Tramonte

 [github.com/LucasTramonte](https://github.com/LucasTramonte)  [Portfolio](#)  [LucasTramonte](#)  [lucas.tramonte@student-cs.fr](mailto:lucas.tramonte@student-cs.fr)

## EXPERIENCE

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### Amazon

Apr. 2025 – Oct. 2025

#### *Business Intelligence Engineer Intern*

- Led research and proof of concept for predicting short-term labor flexibility across Amazon's EU sort center network, tackling complex challenges of temporal dependencies, class imbalance, and facility-specific contexts through SQL data extraction and Python feature engineering.
- Developed ML pipelines in AWS SageMaker Studio using LightGBM and CatBoost models, demonstrating 27.4% improvement over baseline forecasts for same-day predictions through analysis of 3.6 years of historical data across 10 UK sort centers

### FlexSim

Feb. 2023 – Apr. 2023

#### *Data Science Intern - Summer Job*

- Conducted an analysis of possible no-code Machine Learning frameworks, which resulted in RapidMiner being identified as the best viable option due to its ease of use and available features. Instructed the ML engineers of the company on how to use this tool.
- Automated data import in the software, enabling the team to use the software without crashes when handling large datasets, by integrating SQL queries with simulation models using C++.

### Eldorado Institute

Feb. 2022 – Sep. 2022

#### *Data Science Intern in an R&D project funded by Motorola*

- Implemented data pre-processing routines and tested machine learning models to detect faults in mobile phone batteries.
- Improved fault detection precision to 73% by reducing false positives through the use of Python and interpretable machine learning techniques, such as SHAP (SHapley Additive exPlanations).

### Undergraduate research in Machine Learning

Sep. 2022 – Aug. 2023

#### *Funded by National Council for Scientific and Technological Development*

- Exploring SHAP and LIME (Local Interpretable Model-agnostic Explanations) methods for interpreting machine learning predictions in diagnosing SARS-CoV-2 from symptoms - [github](#)

## EDUCATION

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### École CentraleSupélec, Université Paris-Saclay

Aug. 2023 - Oct. 2025

*Master in Engineering Degree. Final year specialization in Artificial Intelligence.*

*GPA: 3.5/4.0*

### State University of Campinas

Mar. 2020 - Jul. 2026

*BS in Industrial Engineering*

*Grade: 9.4/10*

## AWARDS

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### BRAFITEC Excellence Scholarship

Aug. 2023

- Scholarship for a double degree at CentraleSupélec, awarded by the Brazilian government to students with outstanding academic performance.

### International Mathematical Olympics without Borders

Apr. 2019

- My team and I were awarded the bronze medal for national performance at the competition.

## SKILLS

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**Programming Languages:** SQL, Python, C/C++

**Tools and Libraries:** MySQL, AWS, PowerBI, Scikit Learn, Pytorch, Streamlit, Numpy, Pandas, Scipy